

21/09/2026, Kraków

Report of attendance for the training school “Computer Vision in Archaeology” in Brno

Brno is exceptionally well-connected, making it a highly accessible destination to reach for a week of intensive learning. Hosted by the Institute of Archaeology of the Czech Academy of Sciences, the "Computer Vision in Archaeology" training school proved to be a pivotal experience. As a MAIA member specialized in the diagnostics of cultural heritage, I approached this training with a distinct perspective. Stepping into this environment allowed me to bridge my background in physical diagnostics with advanced computational methods, offering a unique opportunity to converse directly with AI specialists and gather fresh, innovative ideas for my own work.

As a member of the MAIA network (COST Action CA23141), this training school offered a pivotal opportunity to become more proactive within the community. While the event was enriched by the presence and interdisciplinary perspectives of members from other Working Groups, it was particularly significant for my own group, Working Group 2 (WG2). WG2 is dedicated to "Digital Comparative Collections and AI Training Data for Archaeology". The group's primary focus is on exploring the availability, accessibility, documentation, and AI-readiness of digital archaeological datasets, which perfectly bridges my background in cultural heritage diagnostics with the computational tools we explored in Brno. A major highlight of the week was meeting Vera Moitinho de Almeida, the leader of WG2. Connecting with her in person allowed me to discuss my research and ideas directly, solidifying my role in the working group and inspiring me to contribute much more actively to MAIA's ongoing initiatives and dataset development.



Technical program

The workshop was structured around real-world archaeological challenges, taking us from introductory concepts to advanced machine learning topics. We kicked off on Monday by setting up our technical environments in Google Colab and diving straight into Python fundamentals for computer vision tasks, including coding with agents.

Throughout the week, the practical sessions were anchored by four primary case studies, allowing us to apply theoretical concepts to varied datasets:

- **Artefacts dataset:** We explored image datasets and annotation theory, tackling the classification and detection of archaeological artifacts from photographs while navigating licensing and controlled vocabularies.
- **Microscopy dataset:** Focusing on microscopic use-wear analysis, we worked extensively on data standardization (brightness, contrast, RGB/grayscale) and used OpenCV for generating detection masks and data augmentation.
- **Satellite imagery & rock art:** Thursday shifted our focus to a macro scale, dealing with remote sensing and object detection on satellite data, alongside rock art analysis.
- **Coins dataset:** The final day brought everything together by combining techniques on photographs and drawings of coins. We explored cutting-edge topics including Large Language Models (LLMs), vision-language models, and the critical issues of model interpretability, bias, attention maps, and causation versus correlation in black-box models.



Showcasing Research: Hyperspectral Reconstruction

A personal highlight of the program was the opportunity to present my own research during the Tuesday afternoon session dedicated to participant projects. I delivered a 15-minute speech focusing on my work in hyperspectral reconstruction. Presenting to this specific cohort was incredibly rewarding; it provided a fantastic platform to disseminate my current research to an audience of potential end-users. The ensuing discussions not only validated the practical utility of the tools I am developing but also provided targeted feedback from peers who understand the intersection of coding and heritage.



Exploring the Institute and Networking

Beyond the theoretical and coding sessions, a highly appreciated addition to the program was the opportunity to visit the institute's laboratories and archive. Seeing firsthand how physical artifacts are stored, curated, and analyzed before digitization provided a tangible link between the material reality of cultural heritage and the digital computer vision workflows we were mastering.

The organizers also struck a perfect balance between rigorous science and enjoyment, integrating convivial moments that made it easy to socialize and establish excellent professional connections. From the ice-breaker event on Monday evening to the Tuesday walking meeting that served as a short city tour of Brno, and the dedicated social event on Wednesday, these informal gatherings were instrumental. They provided a relaxed atmosphere to exchange ideas, debate the day's topics, and build a community with fellow researchers and AI specialists.



Conclusion

The Computer Vision in Archaeology summer school was a remarkably well-executed event that successfully united different disciplines. By combining complex data analysis with a welcoming, highly collaborative atmosphere, it provided exactly the kind of interdisciplinary dialogue needed to push the boundaries of cultural heritage diagnostics.

